Joe Wlos

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PROFESSIONAL EXPERIENCE

Bungalow Taxes: *Co-Founder* **May 2023 – Present**
- Developed React web app for filing residential 'lack of uniformity' property tax appeals in Cook County
- Automated retrieval, cleaning, and analysis of property data to inform appeal recommendations

INTRVL: *Manager, Product and Analytics Engineering* **June 2021 – April 2023**
- Led engineering, creative, and data pipeline development for ad surveys deployed via programmatic advertising networks, text-2-web, and Lucid to measure the effectiveness of liberal groups' online media
  - Captured survey responses from over 90,000 individuals across a dozen client projects
- Visualized COVID-19 vaccine surveys on both internal and public dashboards
  - Built dynamic shareable infographics to display vaccine hesitancy rates at state- and county-levels
  - Finalist for the 2022 Reed Award for "Best Data Visualization"

Hawkfish: *Data Scientist, Voter Research and Business Intelligence* **February 2020 – May 2021**
- Identified and communicated 'red mirage' concept to counter disinformation about vote-by-mail
  - Built online scenario tool–combining state polling, survey data on voting method by candidate choice, and research on state election procedures–to estimate horserace reporting on election night and beyond
  - Conveyed findings to Democratic Party decision makers, media organizations, and the public, resulting in over 2,000 mainstream press reports, 200,000 tweets, and a Shorty Award for "Best Poll or Survey"
- Developed web application for creative teams and marketers to assess second-to-second viewer attention and emotion through Realeyes' pre-testing of video ads, enabling better targeting and content analysis

Dwight Securities Management: *Associate* **June 2017 – October 2019**
- Traded and analyzed CMBS issued by federal agencies and enterprises (Ginnie Mae and Fannie Mae)
  - Built property valuation and cashflow models in Python and VBA to support traders and assess risk
  - Created integrated dashboards for visualizing and searching multifamily and nursing facility datasets

JOOR: *Data Analyst* **June 2016 – May 2017**
- Developed histogram of oriented gradients feature descriptors to classify clothing categories in images
- Used k-means clustering of pixels in product photos to produce visualizations of dominant seasonal colors

Morningstar: *Data Analyst, Reference Data* **July 2015 – May 2016**
- Wrote Python scripts to map identifiers to derivatives using Bloomberg's Open FIGI API

EDUCATION

Grinnell College: *B.A. in Political Science with Honors* **Class of 2015**
- Thesis applying historical development of *habeas corpus* to extrajudicial detentions at Guantánamo Bay
- Winner of 2015 Baumann Essay Prize for research examining the effective regulation of net neutrality

TECHNICAL SKILLS

Data Engineering
- Airflow
- DBT
- SQL

Data Analysis
- Pandas
- NumPy & SciPy
- VBA

Dashboards
- Looker
- Plotly & Dash
- Power BI

Web Development
- FastAPI
- Flask
- React